Color key (phenomenon families)

- State variables $\mathbf{H}, \mathbf{M}, \mathbf{E}, \mathbf{I}, \mathbf{C}$
- Habit & automaticity
- Motivation & reward
- Effort / opportunity cost
- Expectation dynamics
- Identity
- Behavioral complexity
- Fantasy / indulgence
- Collapse / disinhibition
- Context (stress, social, if-then plans)
- Meta-control & gating

$$\theta = C(1 - \mathbf{H}) + \text{stress} \cdot 0.3\,(1 - \mathbf{H}) - ii$$

$$B_{\text{goal}} = \sigma\Big(\frac{\mathbf{M} - \theta}{\tau}\Big)$$

$$B_{\text{habit}} = \mathbf{H}\,\sigma\Big(\frac{\mathbf{M} - M_{\text{habit floor}}}{0.08}\Big)$$

$$\text{blend} = \mathbf{H}^{1.5}, \quad B_{\text{raw}} = (1 - \text{blend})B_{\text{goal}} + \text{blend}\,B_{\text{habit}}$$

$$B = B_{\text{raw}}\,\sigma\Big(\frac{\mathbf{M} - 0.08}{0.03}\Big)$$

Behavioral trigger B. Complexity and stress raise the threshold for goal-directed action; implementation intentions lower it. As $\mathbf{H}$ grows, control shifts from $\mathbf{M}$-vs-threshold toward habitual drive, with a global activation gate.

$$F = f_0 \, \mathbf{M} \, (1 - B)$$

Endogenous fantasy / indulgence. Fires when motivation is present but behavior is not (mental indulgence substitutes for action).

$$\text{decay_rate} = \delta_H \big(1 + 2(1 - B)\mathbf{H}\big)$$

$$\frac{d\mathbf{H}}{dt} = \alpha_H \, B \, \big(A(\mathbf{I}) - \mathbf{H}\big) - \text{decay_rate} \, (1 - B) \, (\mathbf{H} - H_{\min})$$

Habit strength. Asymptotic approach to ceiling $A(\mathbf{I}) = A_{\text{base}} + (1 - A_{\text{base}})\mathbf{I}$ ($identity$ raises the ceiling). Decay accelerates when not practicing, toward residual floor $H_{\min}$.

$$
\begin{aligned}
\frac{d\mathbf{M}}{dt} = & -\beta_M \frac{\mathbf{M}}{1+\mathbf{M}} + \gamma\, B\,(1-\mathbf{H}^2)\,(1-\mathbf{M}/2) - \lambda\,\mathbf{C}\,(1-\mathbf{H})\,B \\
& - \gamma_{\text{ind}}\, F\,(1-\mathbf{H}) - \kappa\,\text{softplus}(-\mathbf{E} - |E_{\text{crit}}|)\,\mathbf{M} - \kappa_{\text{wth}}\,\sigma\!\left(\frac{\mathbf{E} - E_{\text{wth}}}{0.2}\right)\mathbf{M} \\
& + \mu\,(-d\mathbf{E}/dt_{\text{prev}}) + 0.08\max(0,\, 0.2\mathbf{I} - \mathbf{M}) + 0.02\,B\,\mathbf{H} \\
& - 0.3\,\text{stress}\,\mathbf{M}\,(1-\mathbf{H})\,(1 - 0.3\,\text{social}) + 0.05\max(0,\, 0.15\,\text{social} - \mathbf{M}) + \max(0, -d\mathbf{C}/dt)\cdot 0.3
\end{aligned}
$$

Motivational activation. Hyperbolic decay; reward when acting before habit saturates; effort cost rises with complexity when behavior is not automatic. Indulgence drains $\mathbf{M}$. False-hope and "what the hell" collapses remove motivation when $\mathbf{E}$ is extreme. Velocity feedback ties $\mathbf{M}$ to changes in $\mathbf{E}$; identity and habitual maintenance nudge $\mathbf{M}$ upward. Stress and social support modulate drains and sustainment; simplifying behavior ($\mathbf{C}$ falling) gives a small boost.

$$\frac{d\mathbf{E}}{dt} = \phi_M \, \mathbf{M} \, (1 - \mathbf{H}) - \psi \, B \, \mathbf{H} - \epsilon \, \mathbf{E} + \phi_{\text{ind}} \, F$$
$$- 0.3 \, B \, (1 - \mathbf{H}) \, \mathbf{1}_{\{\mathbf{E} > 0\}}$$

Expectation excess. High $\mathbf{M}$ with weak habit inflates $\mathbf{E}$; realized progress and natural recalibration pull it down. Fantasy pushes $\mathbf{E}$ up; early bouts of effort correct inflated expectations.

$$\frac{d\mathbf{I}}{dt} = \alpha_I \, B \, \mathbf{H} \, (1 - \mathbf{I}) - \delta_I \, (1 - B) \, \mathbf{I}$$

Identity coherence. Grows when behavior is habitual; decays slowly when inactive.

$$\mathbf{C}_{\text{equil}} = C_{\text{min}} + (C_{\text{max}} - C_{\text{min}})\,\mathbf{H}\,(0.7 + 0.3\mathbf{I})$$

$$\begin{aligned}\frac{d\mathbf{C}}{dt} = \Big[&-\alpha_{C\downarrow}\max(0, \mathbf{C} - \mathbf{C}_{\text{equil}})\max(0, 0.3 - B)\\ &- \alpha_{C\downarrow}\cdot 0.3\,\text{softplus}\big(-\mathbf{E} - |E_{\text{crit}}|/2\big)\max(0, 0.3 - B)\\ &+ \alpha_{C\uparrow}\max(0, \mathbf{C}_{\text{equil}} - \mathbf{C})\,B\\ &+ \alpha_{C\uparrow}\cdot 0.3\,\mathbf{I}\max(0, \mathbf{H} - 0.3)\max(0, \mathbf{C}_{\text{equil}} - \mathbf{C})\Big]\,(1 + \text{social_support})\end{aligned}$$

Behavioral complexity. $\mathbf{C}$ adapts toward $\mathbf{C}_{\text{equil}}(\mathbf{H}, \mathbf{I})$: performing below one's sustainable complexity enables progressive overload; complexity above equilibrium while B is weak drives simplification; false-hope–type expectation states add extra downward pressure. Social support scales the net adaptation rate.

$$
\begin{aligned}
\text{stress} &\leftarrow \text{stress}\,(1 - 0.1\,\Delta t) \\
ii &\leftarrow ii\,(1 - 0.05\,\Delta t) \\
\text{social_support} &\leftarrow \text{social_support}\,(1 - 0.003\,\Delta t)
\end{aligned}
$$

Transient context variables. Multiplicative decay on different time scales (days to months).